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MIL-P-60463B (PA)
28 October 1974
SUPERSEDING
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14 July 1971

# MILITARY SPECIFICATION

## PROPELLANT POWDER FOR USE IN BASE GRAIN OGK

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 This specification covers Propellant Powder, for use in Base Grain, OGK.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

| | |
|---|---|
| MIL-G-155 - | Graphite (For Use in Ammunition) |
| MIL-C-206 - | Cellulose, Cotton Linters |
| MIL-N-244 - | Nitrocellulose (For Use in Explosives) |
| MIL-N-3399- | 2-Nitrodiphenylamine |
| MIL-L-758 - | Lead Stearate |
| MIL-D-13796 - | Dioctylphthalate |
| MIL-C-13609 - | Casting Solvent for OGK, OIO and AHH Propellants |
| MIL-A-48078 - | Ammunition, Standard Quality Assurance Provisions, General Specification for |

STANDARDS

MILITARY

| | |
|---|---|
| MIL-STD-129 - | Marking for Shipment and Storage |
| MIL-STD-286 - | Propellants, Solid: Sampling, Examination and Testing |

FSC: 1376

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2.2 Other Publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids, or request for proposals, shall apply.

CODE OF FEDERAL REGULATIONS

Title 49, Department of Transportation: Parts 0-199

## 3. REQUIREMENTS

3.1 Materials. The materials shall comply with the requirement of the applicable specification as follows:

| Material | Specifications |
|---|---|
| Nitrocellulose | MIL-N-244, Grade A, Type I Made with Cellulose, Cotton Linters of MIL-C-206 |
| 2-Nitrodiphenylamine | MIL-N-3399 |
| Graphite | MIL-G-155, Grade III or IV |
| Lead Stearate | MIL-L-758 |
| Dioctylphthalate | MIL-D-13796 |

3.2 Composition. The composition shall conform to the requirements of Table I when tested as specified in 4.5.1.

TABLE I

| Material | Requirement (Percent) |
|---|---|
| Nitrocellulose | 88.00 ± 1.25 |
| Dioctylphthalate | 5.00 ± Nominal |
| Lead Stearate | 5.00 + Nominal |
| 2-Nitrodiphenylamine | 2.00 ± .20 |
| Graphite (Glaze) | .03 to .06 |



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3.3 Total Volatiles. The total volatiles shall not be greater than 1.50 percent when determined as specified in 4.5.3.

3.4 134.5°C Heat Test. The powder shall not cause complete fading of the methyl violet paper to salmon pink color in less than 40 minutes and shall not explode in 5 hours when tested as specified in 4.5.4.

3.5 Residual Solvent. The residual solvent shall not be greater than 0.70 max. when determined as specified in 4.5.5.

3.6 Grain Dimensions. Unless otherwise specified in the contract or purchase order, the grain dimensions when determined as specified in 4.5.7 shall be as follows:

| | |
|---|---|
| Length | 0.030 ± 0.003 inches |
| Diameter | 0.030 ± 0.003 inches |

The ends of the grains shall be square and free of chips and tails.

3.7 Packing Density. The packing density of the powder shall be 0.95 min. when determined as specified in 4.5.6.

3.8 Heat of Explosion and Strand Burning Rate on Cast Charges prepared in accordance with paragraph 4.5.2.1

3.8.1 Heat of Explosion. The heat of explosion shall be between 740 and 780 cal/gm when determined as specified in 4.5.2.3.

3.8.2 Strand Burning Rate

3.8.2.1 Burning Rate. The propellant shall burn with the rates specified in Table II when determined as specified in 4.5.2.4.

TABLE II

| Pressure Pounds Per Square Inch | Burning Rate-Inches per second | |
|---|---|---|
| | Lower Limit At Minus 40°F | Upper Limit at Plus 140°F |
| 600 | .197 | .284 |
| 800 | .216 | .311 |
| 1000 | .211 | .302 |
| 1200 | .190 | .280 |
| 1400 | | .288 |
| 1600 | | .318 |
| 1800 | | .337 |
| 2000 | | .379 |



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3.9 Workmanship. The best commerical practices shall be used in the formulation of propellant furnished under this specification, and all other applicable documents. The propellant and its standard ingredients shall be protected from the action of direct sunlight and acid fumes. No reground propellant or nitrocellulose shall be used in the manufacture of propellant procured under this specification unless otherwise specified.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of Inspections. The following type of inspection shall be conducted on this item:

a. Quality Conformance Inspection

4.3 Quality Conformance Inspection

4.3.1 Inspection Lot Formation. See MIL-A-48078

4.4 Examinations and Tests. See MIL-A-48078

4.4.1 Chemical Composition and other Tests of Propellant (3.2 through 3.7)

4.4.1.1 Sampling. Two one pound samples shall be selected at random from the lot of propellant. The samples shall be checked for the requirements specified in paragraph 3.2 through 3.7.

4.4.1.2 Rejection. If either sample fails to meet the requirements specified in paragraph 3.2 through 3.7, the lot of propellant shall be rejected.

4.4.2 Heat of Explosion on Cast Charges. The propellant powder shall be cast into three cast charges in accordance with paragraph 4.5.2.1. A sample shall be selected from each cast charge. The heat of explosion shall be determined on each sample selected from the three cast charges.



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4.4.2.1 Rejection. If one of the samples selected from each of the cast charges fails to meet the requirement of 3.8.1, six additional cast charges shall be cast in accordance with paragraph 4.5.2.1. If any one of the additional cast charge samples fails to meet the requirement of 3.8.1, the lot of propellant powder shall be rejected.

4.4.3 Strand Burning Rate

4.4.3.1 Cast Charges. Thirty-six strands shall be made in accordance with paragraph 4.5.2.4 from each of the three cast charges prepared per paragraph 4.5.2.1. Three strands shall be used to determine the average burning rate at each temperature pressure combination specified in paragraph 4.5.2.4 for each cast charge.

4.4.3.2 Rejection. If any one of the average strand burning rates fails to meet the requirement of 3.8.2.1, six additional charges shall be cast in accordance with paragraph 4.5.2.1. Thirty-six strands shall be made from each of these cast charges. Three strands shall be used to determine the average burning rate at each temperature pressure combination specified in paragraph 4.5.2.4 for each charge. If any one of the average burning rates from each cast charge fails to meet the requirement of 3.8.2.1, the lot of propellant shall be rejected.

4.5 Test Methods and Procedures

4.5.1 Composition

4.5.1.1 Nitrocellulose, Major defect. The nitrocellulose content shall be determined in accordance with Standard MIL-STD-286, Method 209.6 or 209.2.

4.5.1.2 Graphite, Major defect, The graphite content shall be determined in accordance with Standard MIL-STD-286, Method 308.1 or 309.1.

4.5.1.3 2-Nitrodiphenylamine, Major Defect.The 2-nitrodiphenylamine content shall be determined in accordance with Standard MIL-STD-286, Method 218.4.

4.5.1.4 Lead Stearate. The Lead Stearate content shall be determined in accordance with Standard MIL-STD-286 Method 311.1 or 311.2 and the results recorded for information only.

4.5.1.5 Dioctylphthalate. The dioctylphthalate content shall be determined in accordance with Standard MIL-STD-286 Method 222.1 or 226.1 and the results recorded for information only.



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4.5.2 Heat of Explosion and Strand Burning Rate on Cast Charge

4.5.2.1 Cast Charge

4.5.2.1.1 The cast charge shall be made to a minimum of three inches in diameter. Actual size shall depend upon the availability of adequate casting molds. Both inhibited and non-inhibited cast charges are acceptable. However, in either case the outer 1/4 inch peripheral layer of the cast charge shall not be utilized to obtain the required samples.

4.5.2.1.2 The OGK casting solvent composition shall comply with the requirements of specification MIL-C-13609.

4.5.2.1.3 OGK Base Grain Powder Composition shall comply with 3.2.

4.5.2.2 Procedure for preparing the cast charge

4.5.2.2.1 The OGK casting solvent and the OGK base grain powder shall be evaluated at 5 to 10 mm Hg for 8 hours minimum.

4.5.2.2.2 The powder shall be screened into the casting mold. A screening plate shall be placed 4 1/2 inches above the top of casting mold. This screening plate shall have 7/64 inch diameter holes on 5/32" centers. A distributor plate shall be placed 4 1/2 inches above the screening plate. Sufficient holes shall be drilled into the distributor plate so that the base grain powder will be evenly dispersed upon the screening plate when the base grain powder falls from the distributor plate onto the screening plate. The holes in the distributor plate shall be not less than .15 inch in diameter and the ratio between the end area of the cast charge to the total area of the distributor holes shall be 22/1. The powder shall be free to fall from the distributor plate, through the screening plate, and into the casting mold without any obstructions.

4.5.2.2.3 The OGK casting solvent shall be flowed up through the packed casting mold.

4.5.2.2.4 The finished casting shall be cured for a suitable length of time to obtain complete cure at 145°F.

4.5.2.3 Heat of Explosion, Major Defect. The heat of explosion shall be determined in accordance with Method 802.1 of Standard MIL-STD-286.



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4.5.2.4 Strand Burning Rate, Major Defect. The strand burning rate shall be determined in accordance with Method 803.1 of Standard MIL-STD-286. Three strands shall be tested under each pressure and temperature listed below:

| Pressure(psi) At -40°F | Pressure(psi) At 140°F |
|---|---|
| 600 | 600 |
| 800 | 800 |
| 1000 | 1000 |
| 1200 | 1200 |
| | 1400 |
| | 1600 |
| | 1800 |
| | 2000 |

4.5.3 Total Volatiles, Major Defect. The total volatiles shall be determined in accordance with Method 103.3 of Standard MIL-STD-286.

4.5.4 134.5°C Heat Test, Major Defect. The 134.5°C Heat Test shall be determined in accordance with Method 404.1 of Standard MIL-STD-286.

4.5.5 Residual Solvent, Major Defect. The residual solvent content shall be determined in accordance with Method 103.4 or 103.5 of Standard MIL-STD-286 except moisture content shall be determined in accordance with Method 101.3 or with Method 102.1 when using Method 103.4.

4.5.6 Packing Density, Major Defect

4.5.6.1 Apparatus. The following pieces of apparatus shall be required: One 500 ml cylinder, three pieces of numbered sectional tubing, a hopper, distribution plate, screen loading plates, and a brass bar. All of these parts shall be made of normalized brass. Other equipment shall include a brass spillage pan, a 9 inch diameter brass base plate with leveling screws, a small spirit level, a spouted container which will hold approximately 550 grams of powder, and a weighing scale accurate to 0.1 gram. The brass spillage pan shall be of sufficient size to serve as a container for the apparatus and a device for plumbing the assembled apparatus. The cylindrical sections shall be machined to fit one above the other and the screen and distribution plate machined so that they can be inserted between these sections. The plates shall contain holes that are leveled on both sides so that either side will allow powder to flow freely when inserted between the tubes.



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4.5.6.2 Assembly. The apparatus shall be assembled as follows (see figures 1 and 2): The circular base plate shall be placed in the spillage pan. This plate contains a circular area of the same outside diameter as the bottom of the 500 ml cylinder. The 500 ml cylinder shall be seated in the circular area. The small holed loading screen shall be placed on top of tube number 3, and this unit set on top of the 500 ml cylinder. The brass distribution plate shall be placed on tube section number 2 and then tube number 1 shall be added. After the numbered tubes and plates are mounted correctly, the funnel shaped hopper shall be placed into tube number 1. The spillage pan shall be grounded to a cold water pipe with a grounding cable, the spirit level placed on the base plate, and the leveling screws adjusted until the plate is level solid.

4.5.6.3 Conditioning The sample of casting powder shall be conditioned to a constant temperature of 80°F. The casting powder does not have to be conditioned for humidity provided the moisture content of the sample is between 0.1 and 3.0 percent and the test is made in a room having a relative humidity between 30 and 80 percent.

4.5.6.4 Procedure. The empty 500 ml cylinder shall be tared, the weight recorded, and the apparatus assembled as specified in 4.5.6.2. The spouted container shall be tared and approximately 545 gm of powder weighed out. The powder shall be dumped into the hopper in one motion. When all the powder has run through the column, all parts of the apparatus above the 500 ml cylinder shall be removed. This shall be accomplished by holding the bottom cylinder firmly and lifting off the other sections as units. Excess powder shall be struck off by using the brass bar with a 1/4 inch edge held flush with the top of the cylinder and with a flow constant movement across the cylinder top toward the operator. The bar shall not be slanted nor excess powder struck off again. The cylinder shall be removed from the base plate and any powder that may be clinging to the sides of the container, brushed off. The powder and cylinder shall be weighed and the weight recorded. The screen loading density shall be calculated to determine compliance with 3.7 by subtracting the tared weight of the cylinder from the gross weight of the powder and cylinder and multiplying the difference by 0.002.

4.5.7 Grain dimensions, Major Defect. The grain dimensions shall be determined in accordance with Method 504.4 of Standard MIL-STD-286.

## 5. PREPARATION FOR DELIVERY



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5.1 Packing

5.1.1 Packing. The base grain shall be packed in accordance with DOT regulations and the best commercial practice,

5.2 Marking

5.2.1 The containers shall be marked in accordance with DOT regulations and MIL-STD-129.

6. NOTES

6.1 Intended use. OGK casting powder is intended for use in the 155MM Rocket Assisted Projectile (M549) Propellant Grain.

6.2 Ordering data. See MIL-A-48078 (PA).

6.3 Submission of AMC Form 1047. See 6.1.2 of MIL-A-48078 (PA).

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Preparing Activity:
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Project Number: 1376-A034



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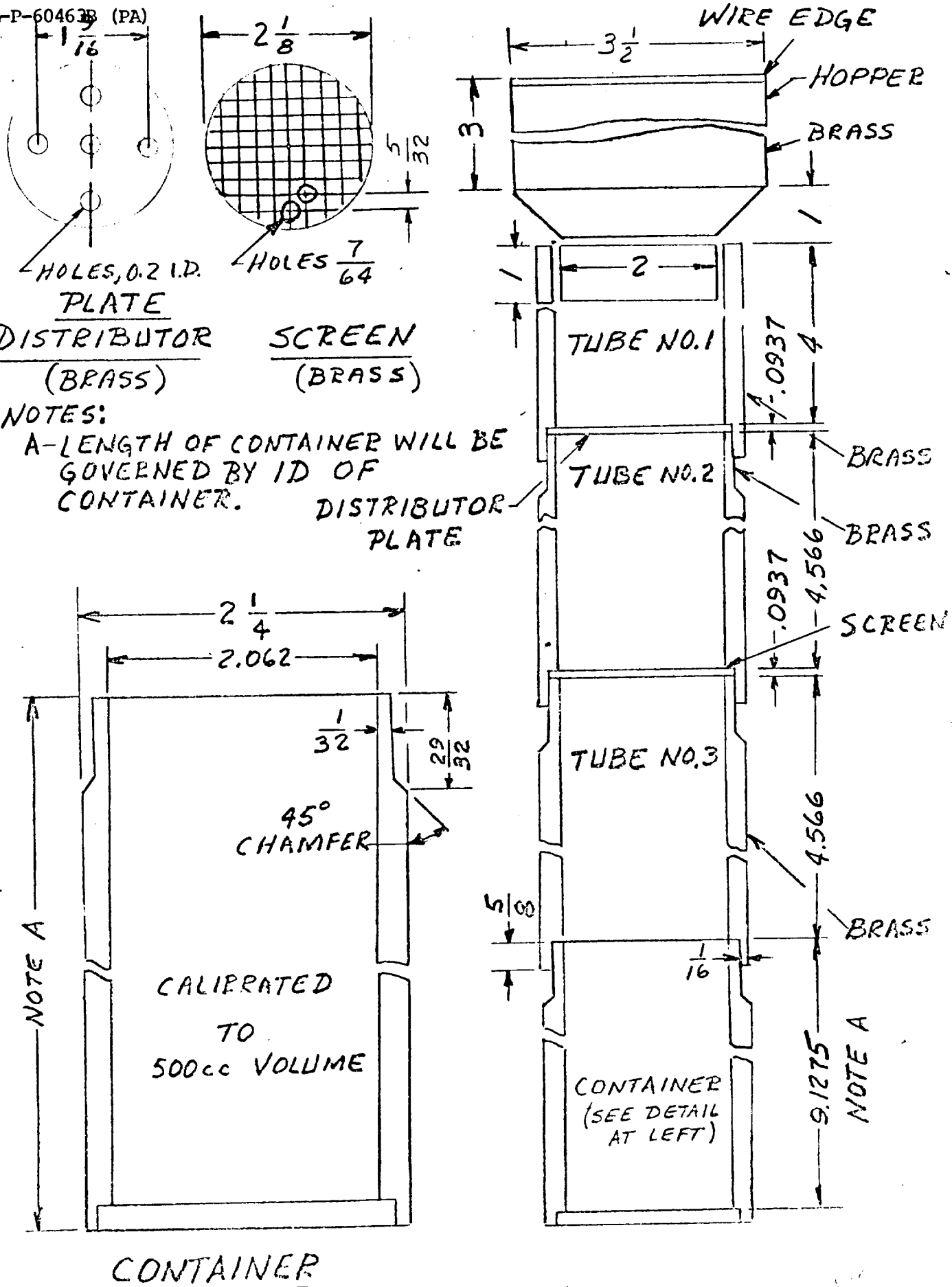



FIGURE 1



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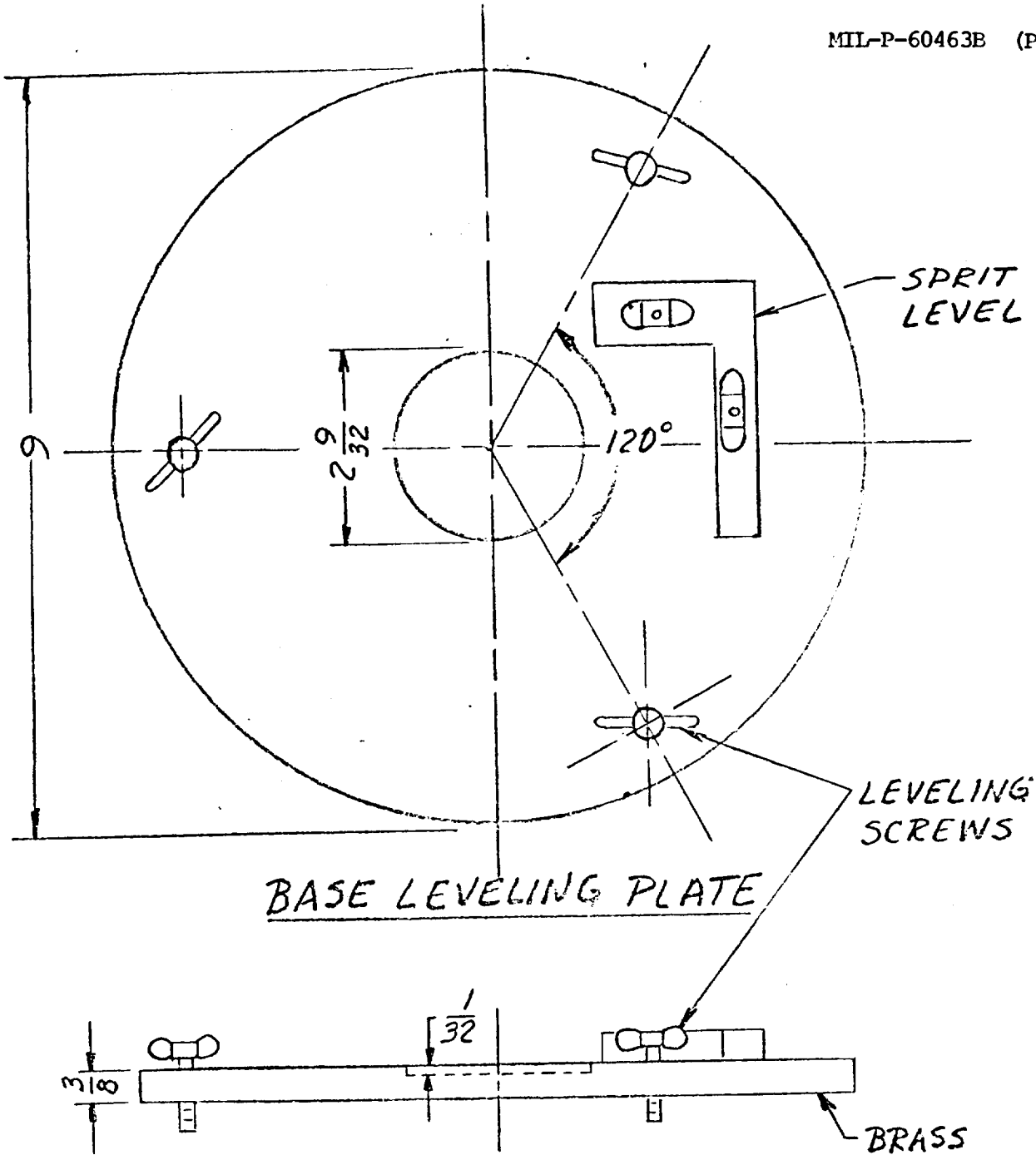



FIGURE 2



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1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

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